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Research Paper

Experimental investigation of synthetic jet–based active cooling for photovoltaic modules: Thermal management, acoustic characterization, and field validation

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ABSTRACT

The electrical efficiency and long-term performance of photovoltaic (PV) modules are adversely affected by heat accumulation within solar cells. Synthetic jet (SJ) actuators have emerged as a promising active cooling technology for PV systems due to their low power consumption and operation without an external fluid supply. However, the influence of nozzle geometry and outdoor performance of synthetic jet cooling on PV modules remain insufficiently investigated. In this study, 57-orifice synthetic jet actuators with nozzle geometries of 3, 5, and 7 mm were experimentally evaluated for PV thermal management. The effects of excitation amplitude, frequency, and orifice diameter on thermal and electrical performance were investigated under 1000 W/m^2 solar irradiance. The optimal conditions were obtained at a 5 mm orifice diameter, 100 Hz frequency, and 3 V amplitude, resulting in a 16.5% reduction in average panel temperature and a 14.9% increase in maximum electrical power output. Outdoor experiments conducted in Erzurum demonstrated a reduction of up to 12 °C in surface temperature and a 7.5% increase in average electrical power output compared to the reference panel. Techno-economic analysis indicated a net annual energy gain of approximately 4.25%; however, the prototype-scale SJ system cost limited improvement in the levelized cost of electricity (LCOE). The obtained results are competitive with water-based active cooling systems, which typically achieve 6–22% power gains but require continuous water supply and maintenance. Overall, synthetic jet-based cooling provides a water-free, low-power, and scalable thermal management solution for photovoltaic systems under realistic operating conditions.

1. Introduction

The rapid increase in global energy demand and the environmental pressures associated with fossil fuel dependence have accelerated the development of renewable energy technologies [1–3]. Among these technologies, solar energy has emerged as one of the most promising alternatives due to its wide availability and environmental advantages [4]. Photovoltaic (PV) systems have gained widespread application owing to their modular structure, scalability, and relatively low operating costs [5,6]. However, the electrical performance of PV systems is highly sensitive not only to incident solar irradiance but also to cell temperature. Only approximately 13–20% of the incident solar radiation is converted into electrical energy, while the remaining energy accumulates within the cells as waste heat [7–10]Consequently, increasing panel temperature significantly reduces electrical efficiency and accelerates long-term module degradation. Previous studies have reported that each 1 °C increase in PV cell temperature results in an approximately 0.3–0.6% decrease in electrical efficiency [11–14]. Particularly under desert and tropical climate conditions, PV panel temperatures may reach 60–70 °C, negatively affecting both instantaneous power generation and long-term system reliability [15–17]. Therefore, effective thermal management of PV modules is essential not only for improving electrical efficiency but also for increasing annual energy production and reducing the levelized cost of electricity (LCOE) [18–20]. The influence of temperature on the electrical performance of PV modules can be explained using the single-diode equivalent circuit model. The current–voltage behavior of a PV module is expressed as;

$$I = I_L - I_s\left[exp\left[\frac{q(V+IR_s)}{AkT}\right]-1\right]-\frac{V+IR_s}{R_{sh}} \quad (1)$$

Where I_L represents the photocurrent, I_S is the diode reverse saturation current, and R_S and R_{Sh} denote the series and shunt resistances, respectively. Due to the exponential dependence of the reverse

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Nomenclature

Abbreviations

LCOE	Levelized Cost of Energy
PV	Photovoltaic
PCM	Phase Change Material
PV/T	Photovoltaic/Thermal
TEG	Thermoelectric Generator
FFT	Fast Fourier Transform
SJ	Synthetic Jet

Symbols

V	Voltage (V)
I	Current (A)
I_L	Photogenerated current (A)
I_S	Diode saturation current (A)
δ	Solar declination angle
μ	Kinematic viscosity (m^2/s)
R_S	Series resistance (Ω)
R_{Sh}	Shunt resistance (Ω)
L_{eq}	Equivalent cavity length (m)
f_H	Resonant frequency (Hz)
A	Surface Area (m^2)
υ	Speed of sound (m/s)
V	Mean velocity (m/s)
T	Temperature (K)
t	Time (min)

saturation current on temperature, an increase in cell temperature reduces the open-circuit voltage and shifts the maximum power point toward lower voltage levels [21,22]. Consequently, temperature rise directly limits the attainable maximum power output of PV panels.

As illustrated in Fig. 1, increasing cell temperature significantly decreases the open-circuit voltage and maximum power output of the PV panel, while the variation in short-circuit current remains relatively limited. Therefore, effective temperature control is crucial for recovering lost electrical power and improving overall system efficiency.

To address this challenge, numerous passive and active cooling techniques have been developed for PV panels. Among passive cooling approaches, phase change materials (PCM), evaporative cooling techniques, hollow-fin and metal-foam-assisted structures, radiative cooling coatings, and hydrogel-based systems have been widely investigated. PCM systems can provide certain temperature reductions; however, their practical applicability is limited by low thermal conductivity and long-term stability issues [23,24]. Hydrogel and evaporative cooling systems offer advantages such as low cost and minimal energy requirement, although their performance strongly depends on ambient humidity and evaporation conditions [17,25–27]. Hollow-fin and metal-foam structures enhance convective heat transfer by increasing the effective surface area, yet manufacturing complexity and limited standalone cooling capability remain important limitations [28–32]. Radiative cooling, on the other hand, strongly depends on atmospheric window conditions [33,34]. Active cooling methods generally provide higher heat transfer coefficients and more effective temperature control. Water-film cooling, serpentine circuits, PV/T integrations, and spray-based cooling techniques have been reported to achieve significant temperature reductions and electrical performance improvements [35–43]. Nevertheless, many active cooling systems require continuous water supply, additional pumping power, and maintenance infrastructure, limiting their applicability in large-scale or water-scarce PV installations [44–50]. Nanofluid-based cooling and hybrid PV/T–TEG approaches may further enhance thermal performance; however, increased pumping power requirements, particle stability issues, and long-term economic uncertainties remain significant challenges for practical implementation [51–60].

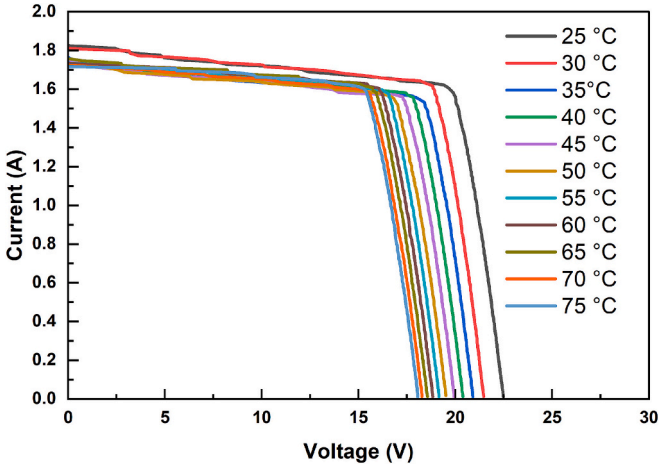


Fig. 1. Effect of temperature increase on the I—V characteristics of a photovoltaic panel.

Nevertheless, for active cooling systems, the achieved temperature reduction must translate into a net energy gain. Therefore, the auxiliary energy consumption of the cooling system should remain sufficiently low to avoid offsetting the recovered electrical power. Accordingly, PV cooling strategies should be evaluated from an integrated energy systems perspective [54]. Existing studies collectively indicate the need for alternative cooling techniques capable of providing high heat transfer performance with low power consumption, without requiring external water supply, while maintaining structural simplicity [9,13,61].

In this context, synthetic jets (SJ) have emerged as a promising active cooling technique due to their compact structure, low power consumption, and operation without external fluid supply [62,63]. In electronic cooling applications, synthetic jets have been reported to provide significantly higher local heat transfer coefficients than natural convection while maintaining low actuator power consumption. In addition, periodic vortex ring formation and boundary-layer re-energization enable enhanced convective heat transfer over extended surfaces, highlighting the scalability potential of SJ-based cooling systems [64]. Despite these promising characteristics, several important gaps remain in the current literature:

1. Geometrical gap: Most synthetic jet studies focus on electronic components or compact heat sinks, while the thermal performance of SJ structures on large-surface PV modules and the systematic influence of nozzle geometry have not been sufficiently investigated.
2. Lack of integrated physical analysis: Existing studies generally examine acoustic behavior, Reynolds number characteristics, and thermal performance separately, whereas integrated analyses linking acoustic excitation, flow regime, and PV electrical performance remain limited.
3. Lack of field validation: The sustainability and repeatability of SJ cooling performance obtained under laboratory conditions have not been sufficiently validated under real outdoor operating environments.

To the best of the authors' knowledge, a comprehensive SJ-based PV cooling framework integrating geometrical optimization, acoustic–flow characterization, and outdoor field validation has not yet been systematically reported in the literature.

The present study addresses these limitations by proposing a synthetic jet-based cooling system that operates without external water supply, requires minimal auxiliary power, and has been validated under real outdoor conditions. Unlike many existing passive or hybrid cooling approaches, the proposed system integrates acoustic characterization, flow regime analysis, and outdoor field validation within a unified experimental framework, thereby strengthening its practical applicability for PV thermal management.

In this study, three nozzle geometries with 57 orifices (3, 5, and 7 mm in diameter) were designed to enhance heat transfer over the PV panel surface. The performance of the synthetic jet system was experimentally investigated by systematically varying excitation amplitude (1–3 V), frequency (50–150 Hz), and orifice diameter. First, panel temperature and maximum power output were measured under 1000 W/m^2 irradiance conditions to determine the optimum operating parameters of the SJ system. Subsequently, FFT-based acoustic pressure spectra and Reynolds number analyses were employed to investigate the relationships among acoustic excitation, flow regime, and cooling performance, thereby enabling the identification of stable operating conditions. Finally, the optimum operating parameters determined under laboratory conditions were experimentally validated through outdoor tests using two identical PV panels, one cooled and the other uncooled. In addition, auxiliary power consumption, exergy efficiency, and levelized cost of electricity (LCOE) analyses were performed to evaluate the thermodynamic and techno-economic feasibility of the proposed cooling approach. Thus, the present study provides a comprehensive assessment of synthetic jet-based PV cooling systems with respect to thermal performance, physical mechanisms, and practical applicability.

2. System description

2.1. Cavity structural design

Plexiglas was selected for the fabrication of the synthetic jet actuator cavity due to its mechanical strength, optical transparency, and ease of machining. The cavity structure consisted of two square plates (250 mm × 250 mm × 3 mm) and four rectangular side plates (250 mm × 40 mm × 3 mm). All components were precisely machined and assembled using appropriate adhesives to ensure dimensional stability and airtight sealing during the experiments.

The transparent structure enabled qualitative visualization of synthetic jet formation and internal flow behavior, facilitating experimental alignment and operational verification. The actuator was successfully operated under both laboratory and real outdoor field conditions in Erzurum, demonstrating stable structural performance under actual environmental conditions. In addition, the modular cavity design enables substituting materials for large-scale or long-term applications without altering the fundamental operating principle. Fig. 2 presents the geometric design of the synthetic jet actuator cavity, including the actual experimental setup and the three-dimensional SolidWorks model [65]. The principal geometric dimensions and manufacturing tolerances of the synthetic jet actuator cavity are summarized in Table 1.

Table 1
Geometric dimensions and tolerances of the synthetic jet actuator.

Component	Dimension	Tolerance
Cavity plate (square)	250 mm × 250 mm	±0.5 mm
Cavity side plate	250 mm × 40 mm	±0.5 mm
Plate thickness	3 mm	±0.1 mm
Orifice diameter (Type A)	3 mm	±0.05 mm
Orifice diameter (Type B)	5 mm	±0.05 mm
Orifice diameter (Type C)	7 mm	±0.05 mm
Circular arrangement diameter	200 mm	±0.5 mm
Number of orifices	57	–
Nozzle-to-surface distance	30 mm	±1 mm

$$L_{eq} = L_n + 0.62D \tag{2}$$

The equivalent neck length L_{eq}, of the Helmholtz resonator is defined by adding an end correction related to the orifice diameter to the physical neck length L_n, as given in Eq. (2).

$$f_H = \frac{v}{2\pi}\sqrt{\frac{A}{V_0 L_{eq}}} \tag{3}$$

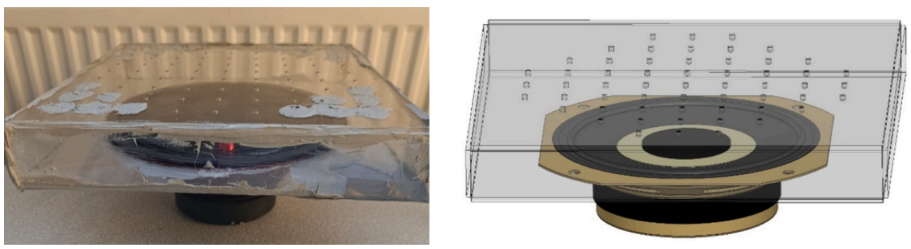
The Helmholtz resonance frequency f_H, is expressed by Eq. (3), where A represents the orifice cross-sectional area, V_0 is the cavity volume, D is the orifice diameter, L_{eq} is the equivalent neck length, and v is the speed of sound = 344 m/s.

The theoretical Helmholtz resonance frequencies were calculated as 302 Hz, 441 Hz, and 555 Hz for the 3 mm, 5 mm, and 7 mm nozzle configurations, respectively. Therefore, the experimental operating frequency range of 50–150 Hz was intentionally selected below the resonance frequencies to ensure stable non-resonant acoustic operation. The Helmholtz resonator model employed in this study assumes: (1) small-amplitude pressure oscillations inside the cavity, (2) incompressible flow within the orifice neck, (3) uniform pressure distribution inside the cavity, and (4) validity of the end correction factor 0.62D for cylindrical orifices. Considering the low excitation amplitudes (1–3 V) and sub-resonance operating frequencies used in this study, these assumptions were considered valid under the experimental conditions.

2.2. Design of orifice geometry

In this study, three different orifice plate configurations were employed, each containing 57 circular orifices. The orifice diameters were systematically varied to investigate the geometric effects on synthetic jet formation. The diameters were selected as 3 mm for Type A, 5 mm for Type B, and 7 mm for Type C. Keeping the number of orifices constant in all configurations enabled the evaluation of the influence of orifice diameter on synthetic jet formation independently.

The orifices were precisely positioned on a circular arrangement with a diameter of approximately 20 cm, ensuring that they remained within the effective vibration area of the loudspeaker diaphragm. This layout ensured that the actuator operated under similar excitation conditions across all configurations, thereby improving the consistency



(a) (b)

Fig. 2. Geometric design of the cavity: (a) real photograph, (b) SolidWorks model.

of flow characteristics and the repeatability of experiments. The systematic selection and positioning of the orifice diameters enabled a detailed investigation of synthetic jet formation and the associated flow behavior under different geometric conditions. The geometries of the orifice plates and their placement within the experimental setup are presented in Fig. 3.

2.3. Experimental setup

In this study, the cooling parameters of the synthetic jet system were first determined under laboratory conditions and subsequently validated under real outdoor environments. The experimental investigation was conducted in two stages: laboratory experiments and field experiments. In both stages, average panel temperature and electrical power output were recorded to evaluate the applicability and performance impact of the optimum operating conditions under practical environmental conditions.

(1. Solar simulator, 2. PV panel, 3. Pyranometer, 4. Thermal camera, 5. Power analyzer, 6. Cavity, 7. Data logger, 8. Spectrum analyzer, 9. Hot-wire probe, 10. Computer, 11. Amplifier, 12. Signal generator)

Fig. 4 illustrates the schematic diagram of the experimental system. The signal generation unit produced and amplified the waveform signals required for synthetic jet operation. An XJX-69 loudspeaker with a diameter of 20 cm was employed as the actuator. A PHILIPS PM 5105 signal generator operating within the frequency range of 50–150 Hz was used in the experiments. The generated signals were amplified using a Guney Electronic amplifier with a 50 W power output and a 12 V DC power supply. This configuration provided stable and adjustable acoustic excitation for synthetic jet formation.

Previous studies have reported that the optimum nozzle-to-surface distance for maximizing the heat transfer coefficient is approximately 30 mm [66]. Accordingly, the distance between the jet exit plane and the panel surface was fixed at 30 mm in the present study. This configuration enabled sufficient development of vortex structures before impingement while minimizing excessive hot-air recirculation near the surface.

Under laboratory conditions, halogen lamps were used to simulate solar irradiance of approximately 1000 W/m^2, as verified with a calibrated pyranometer. Although the spectral distribution of halogen lamps differs from that of natural sunlight, this approach is widely used in PV thermal management studies to evaluate relative thermal and electrical performance under controlled conditions [48]. In the laboratory experiments, the rear surface of the PV panel was directly exposed to the synthetic jet airflow without additional insulation. Ambient temperature was continuously recorded using a calibrated K-type thermocouple positioned adjacent to the panel. The experiments were conducted in an enclosed indoor environment to minimize wind effects during parametric testing. In the field experiments, two identical PV panels were installed side by side under real outdoor conditions. The synthetic jet system actively cooled one panel, while the other panel served as a reference, uncooled. Both panels were exposed to identical environmental conditions, including solar irradiance, ambient temperature, and wind speed, enabling direct performance comparison.

During the experiments, real-time panel temperature data were collected using a FieldLogger data acquisition system. Electrical output was measured using a power analyzer, while acoustic wave

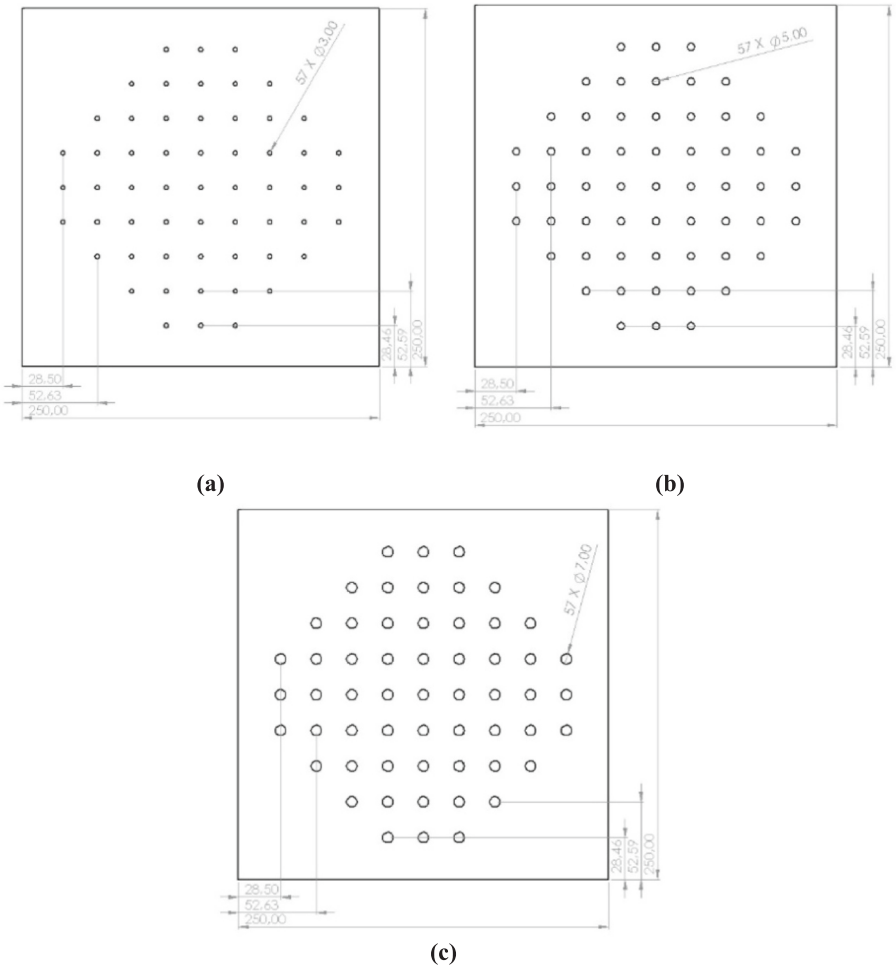


Fig. 3. Nozzle types used in the experimental setup: (a) Type A, (b) Type B, and (c) Type C.

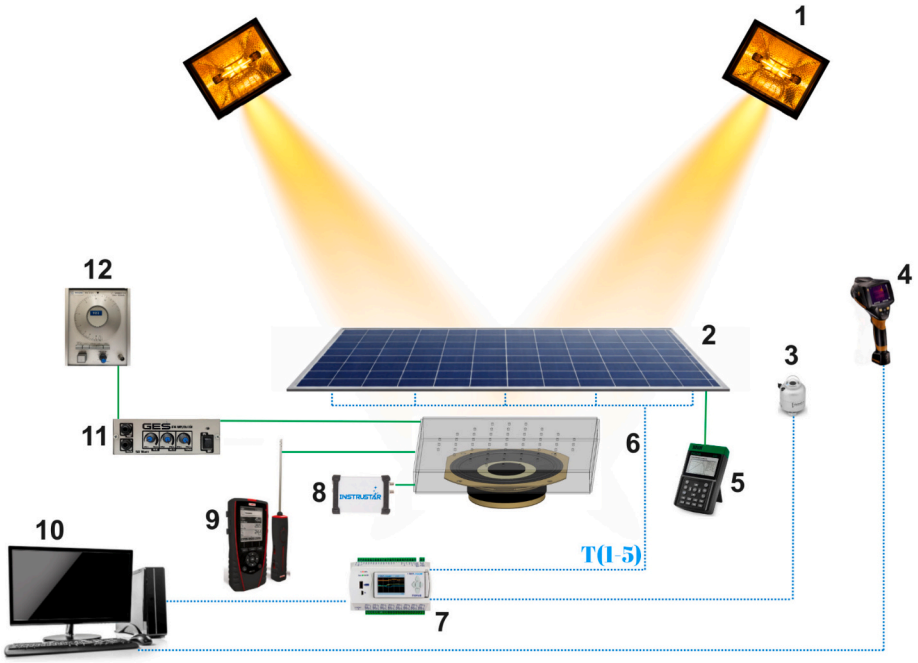


Fig. 4. Schematic view of the experimental system.

characteristics were monitored using an Instrustar spectrum analyzer (ISDS210A). All K-type thermocouples were calibrated against a certified reference thermometer before the experiments, achieving a measurement accuracy of ±0.2 °C. The pyranometer was factory-calibrated and verified against a reference instrument before both laboratory and field deployments. The power analyzer was calibrated according to manufacturer specifications, with voltage and current measurement uncertainties of ±0.03 V and ± 0.03 A, respectively. The surface temperature distribution of the PV panel was analyzed using a Testo-875 thermal camera. The overall views of the experimental setups used in laboratory and field conditions are shown in Fig. 5.

The field experiments were conducted in Erzurum, Türkiye (39.55°N, 41.15°E, altitude 1869 m) under real atmospheric conditions. To represent the outdoor solar conditions, the solar declination angle (δ) was calculated for each experimental day using Eq. (4), where D represents the day number of the year [67]. The PV panels were installed in a south-facing orientation with a tilt angle of 22°, corresponding to the solar geometry of the experimental period. During the experiments, the ambient temperature was approximately 31 °C. According to local meteorological records for the experimental period, the wind speed during midday operating hours (12:00–13:00) was typically 3–4 m/s. These conditions are representative of typical summer operation for PV systems installed in eastern Turkey, where solar irradiance levels frequently approach 1000 W/m^2 during midday.

$$\delta = 23.45^{o} \sin\left[\frac{360^{o}(D+284)}{365}\right] \tag{4}$$

This calculation was used to determine the solar position, while the actual irradiance values were directly measured in the field.



(a) **(b)**

Fig. 5. Experimental setup established for the cooling tests of the photovoltaic panel under laboratory and field conditions: (a) field conditions, (b) laboratory conditions.

2.4. Experimental procedure of the tests

The experimental investigations were conducted in two stages. In the first stage, laboratory experiments were performed under approximately 1000 W/m^2 irradiance conditions. The PV panel was initially operated without cooling until a quasi-steady surface temperature of 75 °C was reached. This temperature level has been reported in the literature as a critical operating threshold at which significant reductions in PV electrical efficiency occur [68]. Subsequently, the synthetic jet actuator was activated, and the excitation amplitude (1–3 V), excitation frequency (50–150 Hz), and orifice diameter (3, 5, and 7 mm) were systematically varied. The nozzle-to-surface distance and jet impingement region were kept constant throughout all experiments to ensure repeatable flow interaction conditions. For each test condition, the time-dependent variations of average panel temperature and maximum electrical power output were simultaneously recorded. Cooling performance was evaluated based on the reduction in average panel temperature and the corresponding increase in electrical power output.

To characterize the flow regime, the Reynolds number at the synthetic jet exit was calculated using Eq. (5):

$$Re = \frac{\rho V D}{\mu} \tag{5}$$

where ρ represents the fluid density, V is the average flow velocity, D is the orifice diameter, and μ is the dynamic viscosity. The average jet velocity was experimentally measured and determined separately for each excitation frequency.

Under laboratory conditions, six 500 W halogen floodlights were used to maintain an irradiance level of approximately 1000 W/m^2 over the PV panel surface. The irradiance distribution was continuously monitored using a pyranometer to ensure stable radiation conditions during all experiments. In the second stage, the optimum operating parameters determined under laboratory conditions were validated through outdoor field experiments using two identical PV panels. One panel was cooled using the synthetic jet system, while the other operated as an uncooled reference panel under identical environmental conditions.

Fig. 6 presents the surface temperature distribution of the PV panel (61 × 53 cm^2) obtained with a thermal camera under 1000 W/m^2 irradiance. The thermal images were used to identify temperature gradients and to determine the locations of thermocouples for representative temperature measurements. Accordingly, five K-type thermocouples (M1–M5) were mounted on the panel's rear surface to provide spatially distributed average temperature measurements throughout the experiments. The average of these five measurements was used to evaluate the PV panel's overall thermal behavior during the cooling tests.

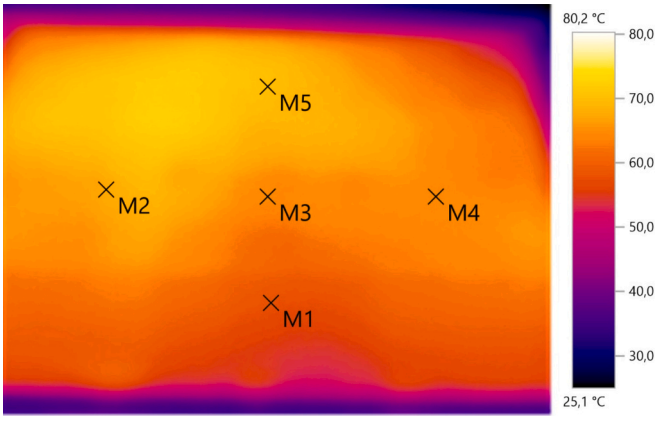


Fig. 6. Surface temperature distribution of the photovoltaic panel under 1000 W/m^2 irradiance.

2.5. Measurement uncertainty and error analysis

The combined standard uncertainty of a result R that depends on several independent input quantities can be evaluated using the law of propagation of uncertainty. In the general case, when the result R is calculated from n input quantities x_1, x_2, ..., x_n with standard uncertainties u_1, u_2, ..., u_n, the combined standard uncertainty u_R is expressed as.

$$u_R = \left[\left(\frac{\partial R}{\partial x_1} u_1\right)^2 + \left(\frac{\partial R}{\partial x_2} u_2\right)^2 + \left(\frac{\partial R}{\partial x_3} u_3\right)^2 + \ldots + \left(\frac{\partial R}{\partial x_n} u_n\right)^2 \right]^{1/2} \tag{6}$$

where u_n is the corresponding standard uncertainty, in this work, Eq. (6) is applied to all calculated performance parameters (such as the average temperature reduction of the PV module, the enhancement in electrical power output, and the Reynolds number of the synthetic jet) to obtain their combined uncertainties. The standard uncertainties u_1, u_2, ..., u_n are determined from the manufacturers' specifications and calibration data of the measurement instruments used in the experiments. All uncertainties are reported in Table 2.

To assess measurement stability, the steady-state temperature values recorded at 20, 25, and 30 min under the optimal operating condition (5 mm orifice, 100 Hz, 3 V) were 63.13 °C, 62.72 °C, and 62.60 °C, respectively, yielding a mean of 62.82 °C with a standard deviation of ±0.28 °C and a coefficient of variation of 0.44%, confirming the repeatability of the experimental measurements.

3. Results

3.1. Thermal performance of the synthetic jet cooling system

Experimental tests conducted at 1000 W/m^2 irradiance without cooling showed a significant decrease in the PV panel's maximum power output as the panel temperature increased. As shown in Fig. 7(a), increasing the panel temperature from 25 °C to 75 °C reduced the maximum power output from 32 W to approximately 24.6 W. Fig. 7(b) presents the time-dependent variation of panel temperature, indicating that the panel temperature reached 75 °C within approximately 13 min under continuous irradiation. These results demonstrate the strong influence of temperature rise on PV electrical performance under uncooled operating conditions.

As illustrated in Fig. 8, experiments conducted at 100 Hz with 5 mm diameter orifices showed that increasing the excitation amplitude improved cooling performance. The average panel temperature decreased by approximately 10.4%, 15.1%, and 16.5% at excitation amplitudes of 1 V, 2 V, and 3 V, respectively. Increasing excitation amplitude enhanced diaphragm displacement, resulting in higher jet exit velocity and momentum flux at the panel surface. Consequently, thermal boundary-layer disruption became more effective, thereby improving convective heat transfer. At 3 V amplitude, the panel

Table 2
Uncertainties of variables.

Variables	Uncertainty (%)
Temperature, T	± 0.2 °C
Frequency, f	± 0.9%
Voltage, V	±0.03 V
Current, I	±0.03 A
Irradiance, W/m^2	± 1.5%
Amplitude, mV	± 0.69%
Power, W	± 2.9%
Re	± 3.2%
Temperature reduction, ΔT	±0.28 °C
Electrical power improvement	±4.1%

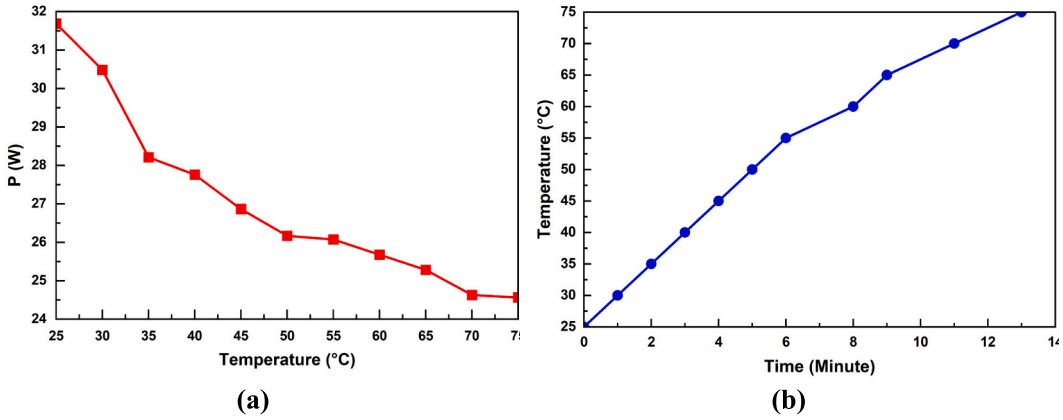


Fig. 7. Variation of power output and temperature of the PV panel under uncooled conditions. (a) Maximum power output versus temperature; (b) Temperature variation with time.

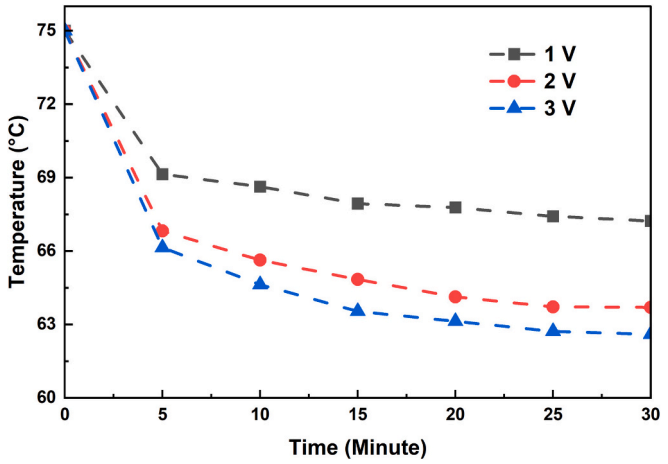


Fig. 8. Time-dependent variation of the average panel temperature at different excitation amplitudes (5 mm orifice diameter, 100 Hz excitation frequency).

temperature decreased rapidly during the initial cooling period and subsequently stabilized at lower equilibrium temperatures compared to the other operating conditions.

The effect of excitation frequency on PV cooling performance was investigated at a constant orifice diameter of 5 mm and an excitation amplitude of 3 V. As shown in Fig. 9, increasing the excitation frequency from 50 Hz to 100 Hz improved cooling performance, whereas further increasing it to 150 Hz reduced cooling effectiveness. The average temperature reduction ratios were 13.5%, 16.5%, and 8% at 50 Hz, 100 Hz, and 150 Hz, respectively. The superior performance at 100 Hz is attributed to the improved balance between jet momentum and periodic boundary-layer renewal. In contrast, lower excitation frequencies reduced the boundary layer disruption frequency, whereas excessively high frequencies weakened the effectiveness of jet-surface interaction.

The influence of orifice diameter on cooling performance was investigated at a constant excitation frequency of 100 Hz and excitation amplitude of 3 V (Fig. 10). After 30 min of operation, the average temperature reduction ratios were measured as 14.8%, 16.5%, and 8.8% for the 3 mm, 5 mm, and 7 mm orifice configurations, respectively. The 5 mm orifice diameter provided the highest cooling performance by achieving a more balanced interaction between jet exit velocity and total momentum flux. Although the 3 mm configuration generated higher exit velocities, the reduced outlet area limited the total momentum transfer. Conversely, the 7 mm configuration produced lower exit velocities, weakening the jet interaction with the panel surface and reducing cooling effectiveness.

3.2. *Electrical performance and field validation of the synthetic jet cooling system*

Under synthetic jet cooling conditions, the maximum electrical

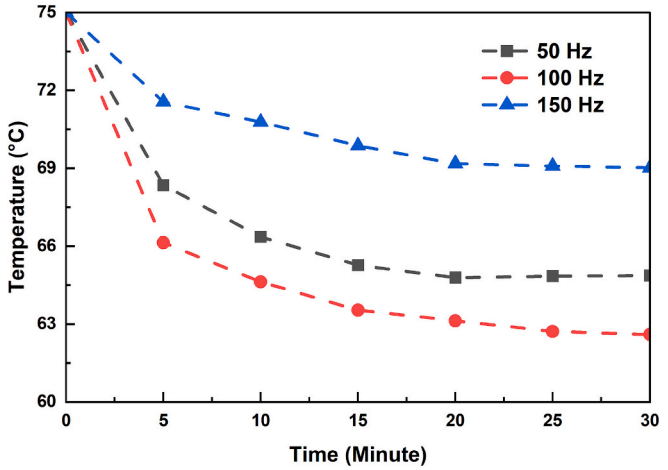


Fig. 9. Effect of excitation frequency on the average temperature of the PV panel (5 mm orifice diameter, 3 V amplitude).

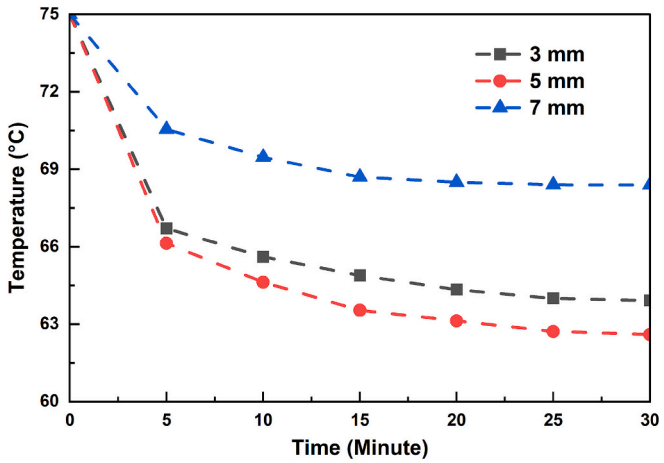


Fig. 10. Effect of orifice diameter on the average temperature of the PV panel (100 Hz excitation frequency, 3 V amplitude).

power output of the PV panel varied significantly depending on the operating parameters (Fig. 11). Increasing the excitation amplitude improved the electrical performance, resulting in power increases of 9.3%, 13.6%, and 14.9% at 1 V, 2 V, and 3 V, respectively. The influence of excitation frequency showed that 100 Hz provided the highest electrical performance, while the improvement at 150 Hz remained comparatively limited. This behavior is attributed to the influence of excitation frequency on jet momentum and surface interaction characteristics. The superior performance at 100 Hz is attributed to the improved balance between jet momentum and the effectiveness of jet-surface interaction. In contrast, lower excitation frequencies reduced the boundary layer disruption frequency, whereas excessively high frequencies weakened the effectiveness of jet-surface interaction. Similarly, the 5 mm orifice configuration produced the highest maximum power output among the 3 mm, 5 mm, and 7 mm configurations. These results indicate that the combination of 3 V amplitude, 100 Hz frequency, and 5 mm orifice diameter provided the optimum electrical performance within the investigated operating range.

The field experiments were conducted under real outdoor conditions, with solar irradiance continuously monitored as a control parameter. The experiments were performed during periods when the irradiance level was approximately 1000 W/m^2, corresponding to the laboratory test conditions. The measured irradiance ranged from approximately 900 to 1100 W/m^2 during the experimental period, as shown in Fig. 12. The results demonstrated that the cooling trends observed under laboratory conditions were also reproduced under outdoor operating conditions.

To evaluate the influence of the synthetic jet cooling system under outdoor conditions, two identical PV panels were initially tested without cooling under an average solar irradiance of 993.2 W/m^2. As shown in Fig. 13(a), increasing the panel temperature resulted in a noticeable reduction in the maximum power output. While the panel produced

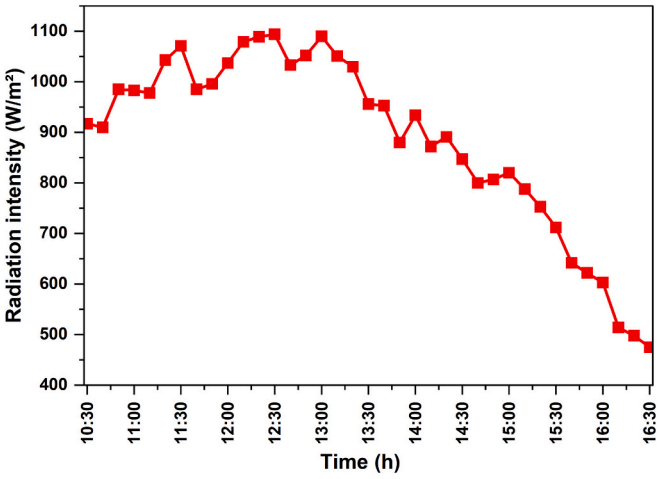


Fig. 12. Time-dependent variation of solar irradiance during the field experiments.

approximately 43.5 W at 35 °C, the power output decreased to 34.8 W at 67 °C, corresponding to an approximate 20% power loss. Fig. 13(b) shows that the panel temperature increased from 35 °C to 67 °C within approximately 40 min under continuous irradiation. These results confirm that elevated panel temperatures significantly reduce PV electrical performance under outdoor operating conditions.

The optimum synthetic jet parameters determined under laboratory conditions (5 mm orifice diameter, 100 Hz frequency, and 3 V amplitude) were applied in the outdoor experiments (Fig. 14). Under an average irradiance of 993.2 W/m^2, the cooled panel temperature decreased from approximately 65 °C to 48 °C within 30 min. In contrast,

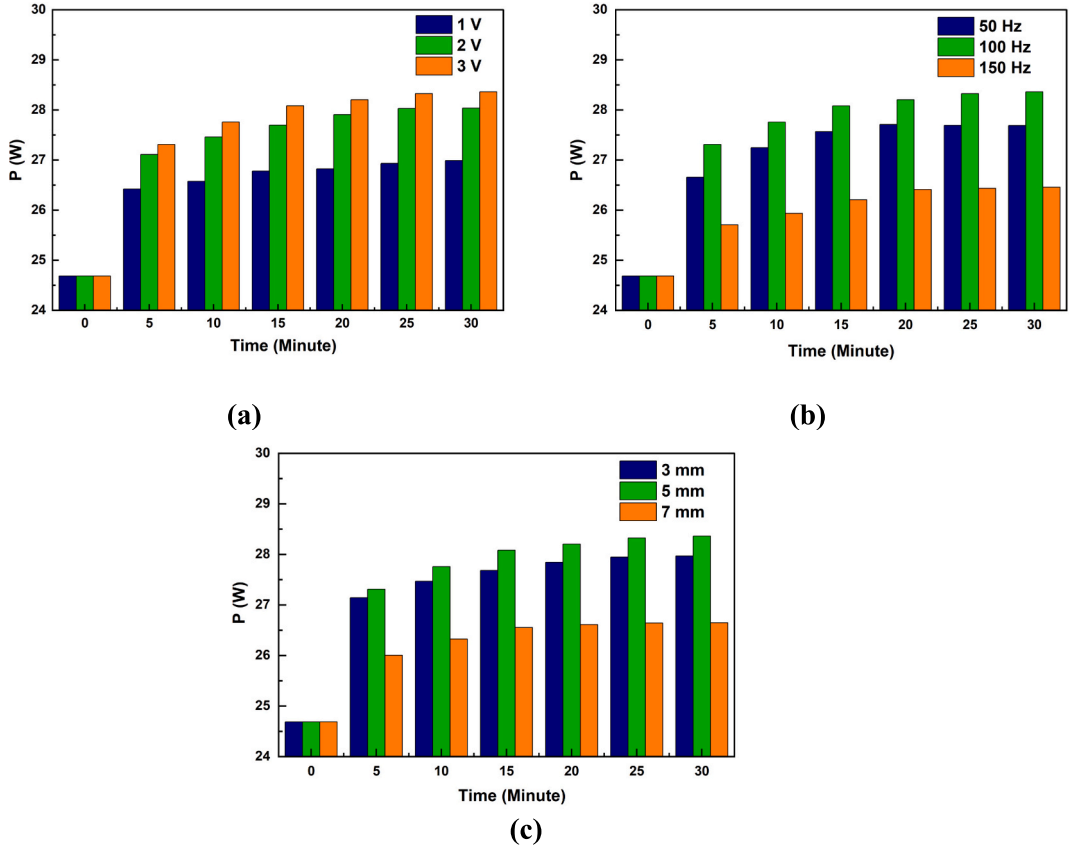


Fig. 11. Effect of cooling parameters on the maximum electrical power output. **(a)** Amplitude; **(b)** Frequency; **(c)** Orifice diameter.

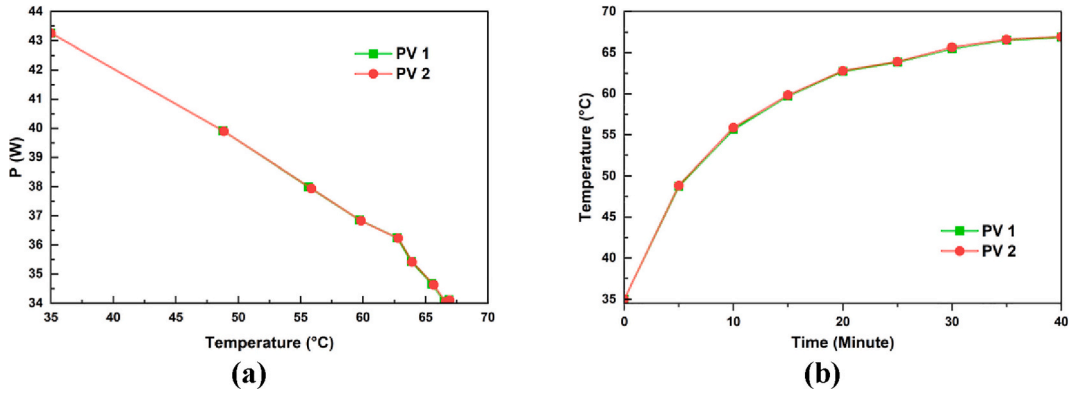


Fig. 13. Thermal behavior and temperature–power relationship of the uncooled PV panel under field conditions. **(a)** Temperature–power characteristics; **(b)** Time-dependent temperature variation.

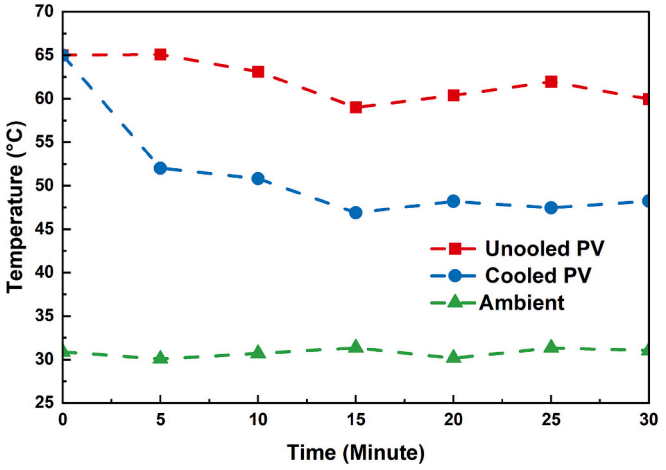


Fig. 14. Thermal performance of the synthetic jet cooling system under field conditions: temperature comparison between cooled and uncooled panels (993.2 W/m^2 irradiance, 5 mm orifice, 100 Hz, 3 V).

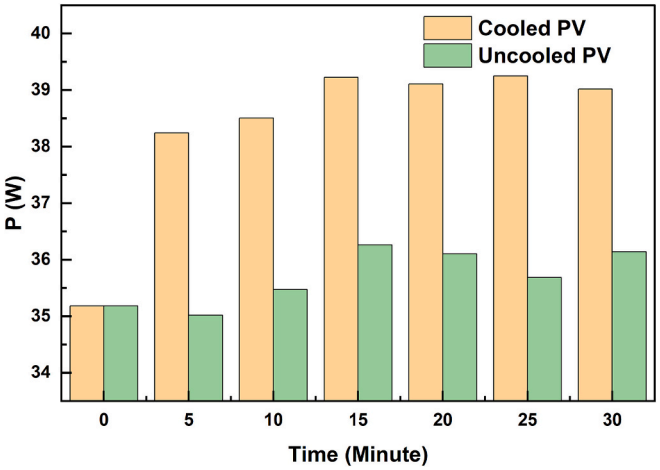


Fig. 15. Comparison of the maximum power output of cooled and uncooled PV panels (993.2 W/m^2 irradiance, 5 mm orifice, 100 Hz, 3 V).

the uncooled panel remained near 60 °C during the same period. Consequently, a temperature difference of approximately 12 °C was obtained between the cooled and uncooled panels. These results demonstrate that the synthetic jet cooling system effectively suppresses panel temperature rise under high irradiance outdoor conditions.

The electrical performance of the cooled and uncooled PV panels was compared under outdoor conditions using the optimum synthetic jet parameters (Fig. 15). At the beginning of the experiment, both panels produced approximately 35 W. After 30 min, the cooled panel reached a maximum power output of approximately 39 W. In contrast, the uncooled panel remained near 36 W. The average power outputs of the cooled and uncooled panels were measured as 38.36 W and 35.69 W, respectively, corresponding to an average electrical power improvement of approximately 7.5%. These findings indicate that reducing panel temperature directly improves PV electrical performance under practical operating conditions.

Table 3 compares the PV cooling methods reported in the literature with the synthetic jet cooling approach proposed in the present study. The synthetic jet system achieved competitive thermal and electrical performance while operating without water consumption and requiring only 2.32 W of auxiliary power. Compared with conventional pump- and compressor-based active cooling systems, the proposed approach provides a low-power and water-free thermal management solution for photovoltaic applications.

Table 3
Comparison of PV cooling methods reported in the literature.

Reference	Cooling Method	Temp. Reduction	Power Improvement	Water Required
Mazzeo et al. [23]	PCM	10–26 °C	Up to 15%	No
Bai et al. [25]	Hydrogel evaporative	21.9 °C	Efficiency: 15.8% → 16.9%	No
Raad et al. [35]	Water cooling	Up to 33 °C	6–20.76%	Yes
Yesildal et al. [5]	Water spray (air-assisted)	~30 °C	Efficiency: 7.9% → 10.2%	Yes
Kilic et al. [48]	Water-acetone spray (air-assisted)	20 °C	–	Yes
This study (Lab)	**Synthetic jet**	**12.4 °C**	**14.9%**	**No**

3.3. *Electrical performance and field validation of the synthetic jet cooling system*

Fig. 16 presents the FFT spectra of the synthetic jet cavity with a 5 mm nozzle under 100 Hz excitation. The broadband spectrum shown in Fig. 16(a) indicates that the dominant acoustic response occurs within the low-frequency region. Therefore, a zoomed spectrum within the 0–500 Hz range is presented in Fig. 16(b) for detailed analysis. A distinct

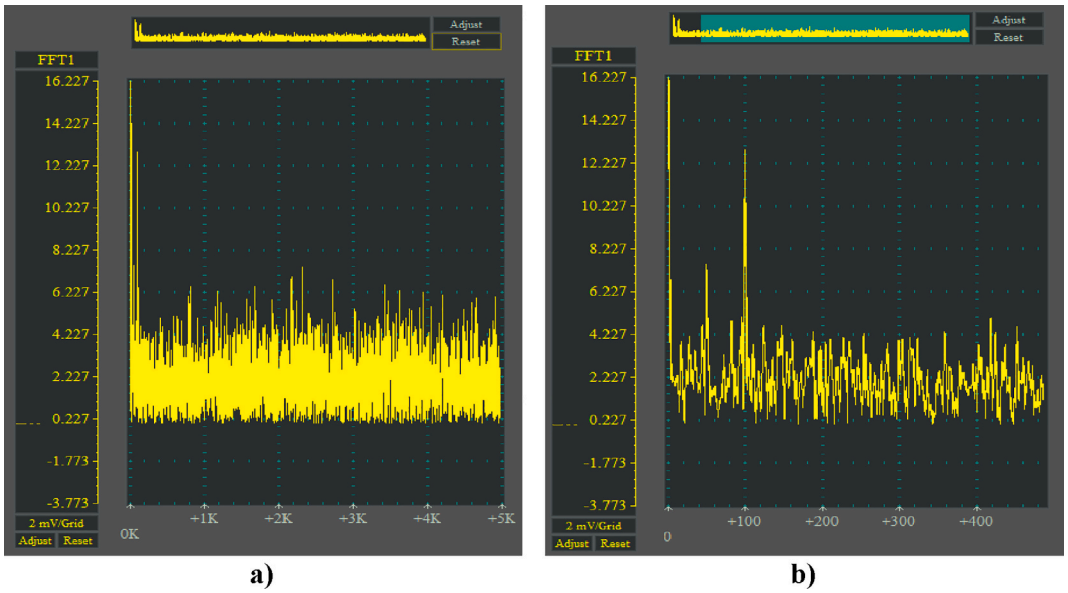


Fig. 16. Acoustic pressure spectrum of the synthetic jet cavity measured from the north direction: (a) broadband FFT spectrum in the range of 0–5 kHz, (b) zoomed spectrum in the 0–500 Hz range where significant responses are concentrated.

dominant peak was observed near 100 Hz, indicating a stable acoustic response associated with the excitation frequency.

Fig. 17 compares the acoustic pressure spectra obtained from four principal directions of the synthetic jet cavity under 100 Hz excitation. A dominant peak near 100 Hz was observed in all directions, confirming that the acoustic response was consistent throughout the cavity. Minor variations in peak amplitude were observed depending on the measurement direction, which may be associated with cavity geometry, sensor positioning, and flow asymmetry. Since the highest signal-to-noise ratio was obtained in the north direction, this orientation was selected as the reference direction for subsequent analyses.

As shown in Fig. 18, the peak acoustic pressure amplitudes increased with increasing excitation frequency for all nozzle configurations. The highest amplitudes were measured with the 7 mm nozzle, whereas the lowest were obtained with the 3 mm nozzle. However, the highest cooling performance was achieved using the 5 mm nozzle configuration. The theoretical Helmholtz resonance frequencies were calculated to be approximately 302 Hz, 441 Hz, and 555 Hz for the 3 mm, 5 mm, and 7 mm nozzles, respectively, indicating that the experimental operating frequencies remained below resonance. These results demonstrate that acoustic pressure amplitude alone does not directly determine thermal performance. Instead, cooling effectiveness depends on the combined influence of jet momentum, flow distribution, and jet-surface interaction characteristics.

Fig. 19 illustrates the variation of Reynolds number with excitation frequency for different nozzle diameters. For all nozzle configurations, the Reynolds number reached its maximum at approximately 100 Hz. The Reynolds numbers measured at 100 Hz were approximately 600, 1050, and 1350 for the 3 mm, 5 mm, and 7 mm nozzles, respectively. Although the 7 mm nozzle produced the highest Reynolds number, the 5 mm nozzle achieved the best cooling performance. This result indicates that thermal performance is not solely determined by the Reynolds number and that flow-surface interaction characteristics play a critical role in synthetic jet cooling performance.

The equivalent neck lengths for the 3 mm, 5 mm, and 7 mm nozzle configurations were calculated as 4.86 mm, 6.10 mm, and 7.34 mm, respectively, using Eq. (2). Although the theoretical Helmholtz resonance frequency increased with orifice diameter, the optimum thermal and electrical performance was obtained with the 5 mm configuration. This result indicates that cooling performance is governed by the balance between jet exit velocity, momentum transfer, and flow–surface interaction characteristics rather than resonance frequency alone. While local heat transfer coefficients were not directly measured, the observed reductions in panel temperature and improvements in electrical output indirectly indicate enhanced convective heat transfer under synthetic jet impingement conditions. Consequently, the improved thermal management achieved with the 5 mm configuration produced the highest electrical performance improvement and yielded a net annual energy gain of approximately 4.25%. However, as discussed in Section 3.4. The prototype-scale cost of the SJ system remained the dominant factor affecting the economic performance, preventing a reduction in LCOE under the assumed cost scenario.

3.4. Performance evaluation and thermodynamic analysis

3.4.1. Power consumption (P_{spent})

The total electrical power consumed to drive the synthetic jet (SJ) system, including the loudspeaker and amplifier, was calculated using the input voltage (V) and the measured current (I):

$$P_{spent} = V.I \tag{7}$$

The low parasitic power consumption observed at the optimum operating condition (3 V excitation amplitude) is due to the compact design of synthetic jet actuators, which do not require an external fluid supply or large-scale mechanical fans.

3.4.2. Thermodynamic assessment: exergy analysis

Beyond conventional energy analysis, a comprehensive exergy analysis was performed to evaluate the system's work potential and thermodynamic quality. In photovoltaic systems, an increase in cell temperature promotes entropy generation, thereby directly limiting both the exergy efficiency and the electrical work potential of the system. In this study, the total solar exergy input to the panel $Ex_{in,sun}$ was calculated using the Petela equation, based on the irradiance intensity G, panel area A, ambient temperature T_a and the effective surface temperature of the sun T_s:

$$Ex_{in,sun} = G.A.\left[1 - \frac{4}{3}\left(\frac{T_a}{T_s}\right) + \frac{1}{3}\left(\frac{T_a}{T_s}\right)^4\right] \tag{8}$$

The net electrical exergy output of the system was obtained by subtracting the auxiliary power consumption of the synthetic jet system

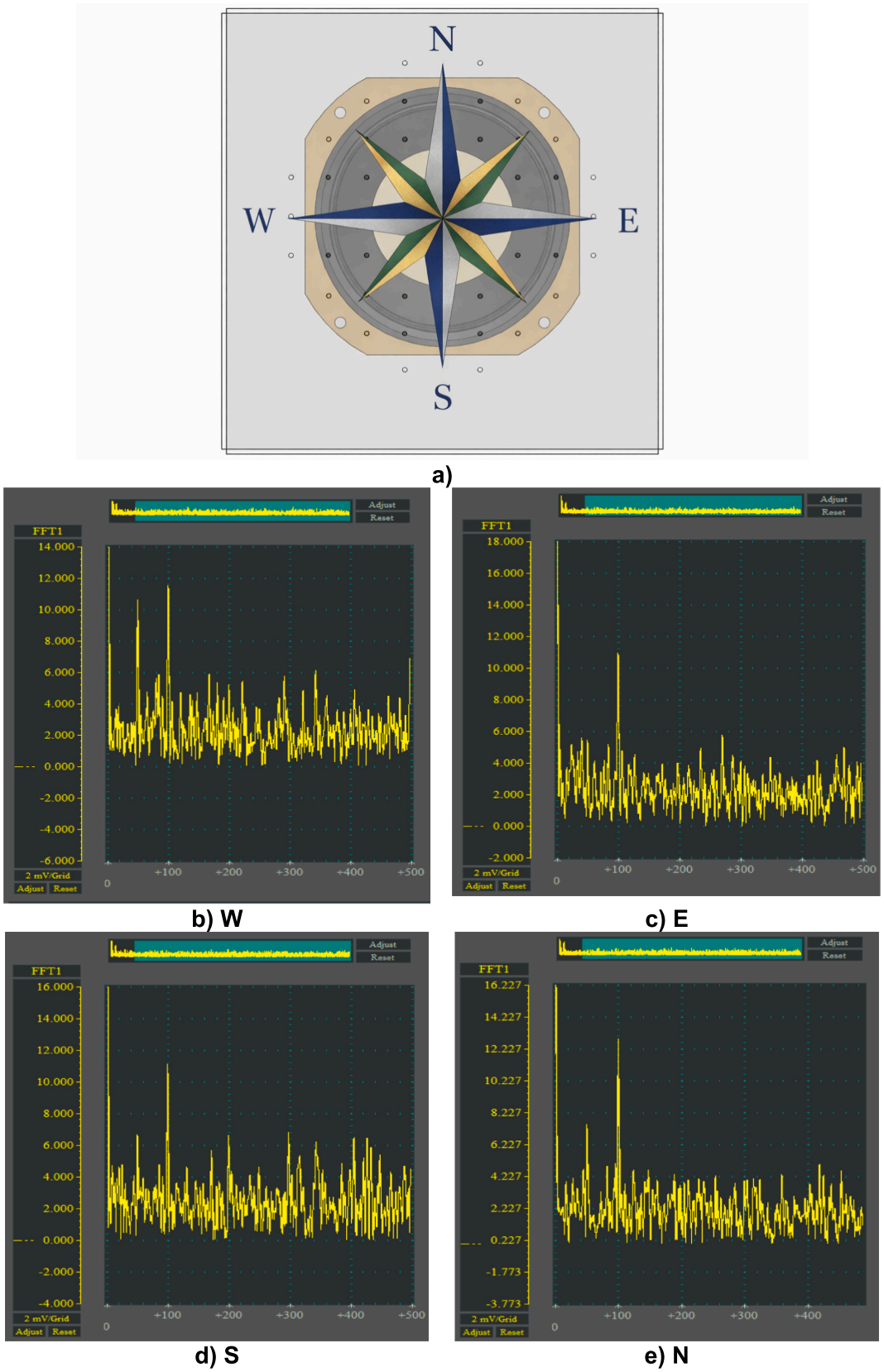


Fig. 17. Acoustic pressure spectra measured from different directions of the synthetic jet cavity with a 5 mm nozzle under 100 Hz pulsed excitation: (a) directional reference, (b) west, (c) east, (d) south, and (e) north directions (FFT results within the 0–500 Hz range).

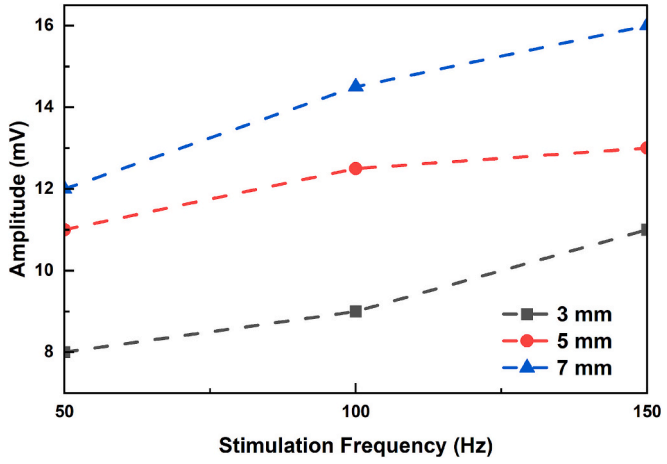


Fig. 18. Variation of peak pressure amplitudes with excitation frequency for different nozzle diameters.

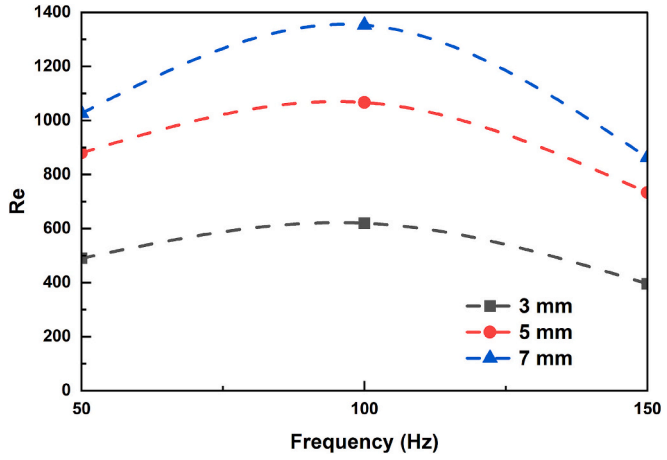


Fig. 19. Variation of Reynolds number with excitation frequency for different nozzle diameters.

from the maximum electrical power generated by the PV panel:

$$Ex_{out,net} = P_{max} - P_{spent} \tag{9}$$

Under field-test conditions in Erzurum, with an average irradiance of 993.2 W/m^2 and an ambient temperature of 304.15 K, the total solar exergy input to the system was calculated to be 298.63 W, considering the effective PV panel area. The net electrical exergy output $Ex_{out,net}$ was determined by subtracting the auxiliary power consumption required to operate the SJ system from the maximum electrical power produced by the PV panel. The overall exergy efficiency was then defined as $\eta_{ex} = Ex_{out,net}/Ex_{in,sun}$.

The numerical analysis indicates that reducing the panel temperature from 333.15 K (60 °C) to 321.15 K (48 °C) results in a measurable improvement in the thermodynamic quality of the system. Under the uncooled reference condition, the exergy efficiency was 11.95%; under cooled conditions, despite the auxiliary energy consumption of the SJ system, it increased to 12.13%. Although the absolute increase of 0.18% appears relatively small, it indicates that the synthetic jet–based active cooling system improves the net thermodynamic performance of the PV system even when the auxiliary energy consumption is considered. In particular, the improvement observed at 100 Hz excitation frequency suggests that the thermally induced exergy destruction within the PV cells is reduced and that the conversion of solar energy into high-quality electrical work is better preserved. The combined uncertainty in exergy efficiency was calculated using the law of propagation of uncertainty, yielding a ± 0.35% uncertainty. While the absolute improvement of 0.18% falls within this uncertainty range, the analysis confirms the energy-positive operation of the synthetic jet cooling system.

3.4.3. Techno-economic and LCOE analysis

The commercial potential of the proposed synthetic jet (SJ) cooling system was evaluated using the Levelized Cost of Electricity (LCOE) methodology. The system boundary included the PV module and the SJ cooling system components (loudspeaker, amplifier, signal generator, and actuator housing). Balance-of-system components such as the inverter, support structure, and electrical infrastructure were excluded because they were identical for both the cooled and uncooled PV systems. The economic assessment was performed for Erzurum, Türkiye, assuming 1600 equivalent full-load operating hours per year, a system lifetime of 25 years, a discount rate of 8%, and annual operation and maintenance costs equal to 2% of the initial investment cost.

The LCOE is defined as the ratio of the total discounted lifetime costs of the system to the total discounted energy production [18]:

$$LCOE = \frac{\sum_{t=0}^{n} \frac{I_t + M_t}{(1+r)^t}}{\sum_{t=1}^{n} \frac{E_t}{(1+r)^t}} \tag{10}$$

where (n) is the system lifetime, (r) is the discount rate, (I_t) is the investment cost, (M_t) is the annual operation and maintenance cost, and (E_t) is the annual electricity production. Annual operation and maintenance costs were assumed to be 2% of the initial investment cost.

Based on experimentally measured average power outputs, the reference PV system was estimated to generate 57.10 kWh/year, while the SJ-assisted system produced 61.38 kWh/year. After accounting for the auxiliary energy consumption of the SJ cooling system, the net annual electricity generation was calculated as 59.53 kWh/year, corresponding to a net energy gain of approximately 4.25%. The cost of the 50 W PV module used in the experiments was approximately 51 USD; however, since the same module was used in both the cooled and uncooled configurations, the economic assessment considered only the additional cost associated with the SJ cooling system. The prototype-scale cost model of the SJ cooling system is summarized in Table 4.

Under the assumed prototype-scale cost scenario, the additional investment cost of the SJ cooling system outweighed the economic benefit from the measured energy gain. Therefore no reduction in LCOE was achieved.

To evaluate the influence of cost uncertainty, a sensitivity analysis was performed by varying the SJ system cost by ±50% relative to the base-case value of 80 USD. The results are presented in Table 5.

The results indicate that the economic performance of the proposed cooling system is highly sensitive to the initial investment cost. Although the SJ-assisted system increased the net annual electricity production by approximately 4.25%, the prototype-scale component cost remained the dominant factor influencing the calculated LCOE. Therefore, the present economic assessment should be interpreted within the context of prototype-scale cost assumptions. Nevertheless, the SJ system costs considered in the present study are based on prototype-scale components used for laboratory implementation. In commercial applications, component optimization, mass production, and economies of scale may reduce these costs, potentially improving the economic feasibility of the

Table 4
Prototype-scale SJ system cost model.

Component	Cost (USD)
Loudspeaker	26
Actuator housing	22
Signal generator	13
Amplifier	19
Total	80

Table 5
Sensitivity analysis of the SJ system cost on LCOE.

SJ System Cost (USD)	LCOE (USD/kWh)
40	0.174
60	0.212
80	0.250
100	0.289
120	0.327

proposed cooling approach.

4. Conclusion

This study experimentally investigated the thermal and electrical performance of a 57-orifice synthetic jet (SJ)-based active cooling system developed for photovoltaic (PV) thermal management under both laboratory and outdoor operating conditions. The results demonstrated that synthetic jet cooling effectively suppresses panel temperature rise and improves PV electrical performance while maintaining low auxiliary power consumption.

- The experimental investigations identified the optimum operating condition as a 5 mm orifice diameter, 100 Hz excitation frequency, and 3 V excitation amplitude, which provided the most effective thermal and electrical performance among the tested configurations.
- Under laboratory conditions at an irradiance level of 1000 W/m^2, the optimized configuration achieved a 16.5% reduction in average panel temperature and a 14.9% increase in maximum electrical power output.
- Outdoor field experiments conducted in Erzurum, Türkiye, under an average irradiance of 993.2 W/m^2 demonstrated a sustained temperature reduction of approximately 12 °C between the cooled and uncooled PV panels.
- Under these conditions, the cooled PV panel produced an average power output of 38.36 W, whereas the uncooled panel generated 35.69 W, corresponding to an average electrical power improvement of approximately 7.5%.
- Acoustic analysis and Reynolds number evaluation showed that the SJ system exhibited stable acoustic behavior and maximum jet generation efficiency near 100 Hz excitation frequency. The results further indicated that cooling performance depends not only on Reynolds number but also on the combined effects of jet momentum, vortex ring formation, and jet-surface interaction characteristics.
- The power consumption of the synthetic jet (SJ) actuator remained lower than the net electrical gain provided by the system, resulting in an approximately 4.25% increase in annual net electricity production. The exergy analysis indicated that the thermodynamic performance of the PV system was maintained under cooled conditions. Although no reduction in LCOE was achieved under the prototype-scale cost assumptions, the proposed cooling approach shows potential for improved economic performance under commercial-scale implementation and reduced component costs.

Overall, the proposed SJ cooling system, characterized by water-free operation, compact structure, and low parasitic power consumption, represents a promising thermal management approach for photovoltaic applications. The system's water-free, low-power operation makes it particularly suitable for large-scale rooftop PV installations in regions with moderate ambient temperatures and limited water resources. In addition, the modular and scalable structure of the synthetic jet actuator allows potential integration into existing PV installations without significant structural modifications. The demonstrated applicability under real outdoor conditions in Erzurum further confirms the practical viability of the proposed approach. Future studies may focus on detailed flow diagnostics and high-resolution numerical simulations to further investigate jet–surface interaction mechanisms in synthetic jet cooling systems. Advanced experimental techniques, such as Particle Image Velocimetry (PIV) and computational fluid dynamics (CFD) modeling, could be employed to directly investigate vortex formation, jet interactions, and periodic vortical structures generated by multi-orifice synthetic jet arrays. In addition, future investigations may evaluate the scalability of multi-orifice synthetic jet arrays for large-scale PV applications and assess system performance under higher ambient temperatures typical of hot, arid climates.

CRediT authorship contribution statement

Hassen Ghaly: Writing – review & editing, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Kenan Yakut:** Validation, Supervision, Project administration, Conceptualization. **Muhammet Harun Osta:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] M. Demir, Optimizing solar water heating systems for carbon emission reduction in residential buildings: a case study of Turkey's renewable energy potential, Case Stud. Therm. Eng. 69 (2025), https://doi.org/10.1016/j.csite.2025.105964.
[2] M. Mustafaoglu, M.K. Yeşiyurt, I. Kotcioglu, A. Razmi, Evaluation of hydraulic and thermal performance in a novel fractal heat exchanger: a comparative study, J. Braz. Soc. Mech. Sci. Eng. 47 (4) (2025), https://doi.org/10.1007/s40430-025-05460-w.
[3] M.H. Osta, K. Yakut, H. Köksal, Experimental performance assessment of two-stage thermoelectric system in a closed-cycle drying process, Appl. Therm. Eng. (2025) 127451, https://doi.org/10.1016/j.applthermaleng.2025.127451.
[4] A.N. Özakin, F. Yeşildal, İ.V. Öner, A.A.H. Al Agele, S.M.H. Baamel, Experimental and numerical thermal analysis of conical solar collector using helical absorber tube, J. Therm. Anal. Calorim. 150 (16) (2025) 12699–12715, https://doi.org/10.1007/s10973-025-14519-8.
[5] F. Yesildal, A.N. Ozakin, K. Yakut, Optimization of operational parameters for a photovoltaic panel cooled by spray cooling, Eng. Sci. Technol. Int. J. 25 (2022) 100983, https://doi.org/10.1016/J.JESTCH.2021.04.002.
[6] M.H. Osta, K. Yakut, A.N. Özakın, F. Yeşildal, Experimental investigation of spray cooling in PV panel using air-assisted nozzle, J. Enhanced Heat Transfer 33 (1) (2026) 1–23, https://doi.org/10.1615/JEnhHeatTransf.2025059896.
[7] A. Alqatamin, S. Jinzhan, Numerical analysis and design of photovoltaic-thermal (PVT) system with novel water-cooling channel structure integrated with perforated V-shape fins, Renew. Energy 243 (2025), https://doi.org/10.1016/j.renene.2025.122587.
[8] S. Chan, et al., Comparative analysis of switchable natural cooling methods for efficient thermal management in photovoltaic/thermal collectors, Appl. Energy 385 (2025), https://doi.org/10.1016/j.apenergy.2025.125499.
[9] M.S. Tahir, X. Dong, M.M. Khan, I.U. Rehman, Thermal management of photovoltaic systems: a comprehensive review of cooling strategies, Elsevier Ltd, 2025, https://doi.org/10.1016/j.solener.2025.113811.
[10] M.H. Osta, K. Yakut, Experimental and numerical analysis of spray cooling in photovoltaic systems, Heat Transf. Res. 56 (12) (2025) 81–94, https://doi.org/10.1615/HEATTRANSRES.2025058227.
[11] S. Aït Saada, R. Nebbali, I. Kecili, D. Zembri-Nebbali, M. Rahim, D.M. Nguyen, Operating temperature and electrical efficiency of a photovoltaic panel cooled by laminar water film flowing on its front face: experimental and numerical approach, Mater. Today Sustain. 30 (2025), https://doi.org/10.1016/j.mtsust.2025.101093.
[12] B. Sutanto, H. Iacovides, A. Nasser, A. Cioncolini, I. Afgan, Efficiency improvement of floating photovoltaic panels with natural convection cooling loops: multi-

physics thermal modelling, Sol. Energy 286 (2025), https://doi.org/10.1016/j.solener.2024.113170.
[13] H. Raad, S. Ali, J. Faraj, C. Castelain, K. Chahine, M. Khaled, Enhancing photovoltaic performance through water-based cooling: a comprehensive review of passive and active techniques, Appl. Therm. Eng. 283 (2026) 128943, https://doi.org/10.1016/j.applthermaleng.2025.128943.
[14] F. Bayrak, A. Gönül, M. Camci, Thermal management of photovoltaic panels using configurations of spray cooling systems, Appl. Therm. Eng. 274 (2025), https://doi.org/10.1016/j.applthermaleng.2025.126656.
[15] H. Dirawi, Q. Wang, M. Hu, Y. Su, S. Riffat, Comprehensive performance analysis of a novel closed-loop hydronic cooling of photovoltaic panel with a controlled intermittent flow strategy, Renew. Energy 239 (2025), https://doi.org/10.1016/j.renene.2024.122185.
[16] A.A. Abdirahman, M. Asif, E. Cuce, I. Ahmed, M. Alqahtani, M. Khalid, Optimizing PV systems in high-temperature environments: a SWOT-based analysis of cooling technologies, Elsevier Ltd, 2025, https://doi.org/10.1016/j.esr.2025.101828.
[17] L. Liu, X. Liu, Q. Zhu, J. He, Research and application of an efficient hygroscopic hydrogel for cooling of photovoltaic panels, Thermal Sci. Eng Progress 68 (2025), https://doi.org/10.1016/j.tsep.2025.104212.
[18] P. Benalcazar, A. Komorowska, J. Kamiński, A GIS-based method for assessing the economics of utility-scale photovoltaic systems, Appl. Energy 353 (2024), https://doi.org/10.1016/j.apenergy.2023.122044.
[19] C. Wang, H. Chen, Z. Wang, Y. Gao, G. Zhang, Economic assessment and grid parity analysis of photovoltaic power generation considering tradable green certificate, Appl. Energy 382 (2025), https://doi.org/10.1016/j.apenergy.2025.125282.
[20] A. Kazemian, H. Yang, C. Xiang, Assessing the effects of photovoltaic and solar thermal ratios on performance, cost, and emissions in combined solar configurations, Appl. Energy 384 (2025), https://doi.org/10.1016/j.apenergy.2025.125438.
[21] A. Kabakuş, A. Öztürk, F. Sönmez, A new approach to cooling photovoltaic panels: electrospray cooling, Case Stud. Therm. Eng. 64 (2024) 105545, https://doi.org/10.1016/J.CSITE.2024.105545.
[22] T. Li, et al., Comparison of the thermoelectric performance of different photovoltaic/thermal hybrid thermoelectric generation modules: an experimental study, Appl. Energy 378 (2025), https://doi.org/10.1016/j.apenergy.2024.124771.
[23] D. Mazzeo, L.P.M. Colombo, S. Leva, A review of photovoltaic cooling with phase change materials: technical advances, modeling approaches, efficiency gains and economic/environmental impact, Elsevier Ltd, 2025, https://doi.org/10.1016/j.applthermaleng.2025.128222.
[24] A. Shahsavar, M. Moradvandi, N. Azimi, Experimental study on ultrasonic-assisted PCM cooling: a novel approach to photovoltaic panel performance improvement, Appl. Therm. Eng. 265 (2025), https://doi.org/10.1016/j.applthermaleng.2025.125642.
[25] B.L. Bai, S. Du, M.J. Li, Photovoltaic efficiency improved by self-adaptive water uptake hydrogel evaporative cooling, Appl. Energy 383 (2025), https://doi.org/10.1016/j.apenergy.2025.125367.
[26] Y. Ding, et al., Boosting photovoltaic efficiency via efficient passive cooling from all-natural hydrogels with robust interfacial adhesion, J. Energy Chem. 114 (2026) 206–216, https://doi.org/10.1016/j.jechem.2025.09.089.
[27] M.M. Abdel-Aziz, A.A. ElBahloul, Optimization of different composite sorption materials and their thickness for enhanced PV cooling performance: a multiphysics simulation approach, Sol. Energy Mater. Sol. Cells 285 (2025), https://doi.org/10.1016/j.solmat.2025.113554.
[28] Y. Wang, L. Pan, J. Hao, D. Ding, K. Tan, Z. Zhao, Performance investigation of a novel passive cooling strategy for photovoltaic panels using hollow staggered triangular fins, Case Stud. Therm. Eng. 76 (2025) 107326, https://doi.org/10.1016/j.csite.2025.107326.
[29] M. Sheikholeslami, M.K.A. Al-Bujasim, Efficiency enhancement of photovoltaic solar panels using hybrid cooling systems with cross-shaped fins and porous foam, Int. Commun. Heat Mass Transf. 167 (2025), https://doi.org/10.1016/j.icheatmasstransfer.2025.109331.
[30] K. Sopian, et al., Performance analysis of photovoltaic panels with dual-axis tracking and passive cooling using fins and radiative paint, Sol. Energy 302 (2025), https://doi.org/10.1016/j.solener.2025.114047.
[31] F.T. Bayrak, B.İ. Toprak, İ. Solmaz, Ö. Bayer, The influence of slotted solid cylindrical fin with aluminum foam insert on the thermal and hydraulic characteristics of air-cooled pin-fin heat sink, Int. Commun. Heat Mass Transf. 162 (2025) 108595, https://doi.org/10.1016/J.ICHEATMASSTRANSFER.2025.108595.
[32] M.T. Topcu, B.İ. Toprak, K. Bilen, Experimental and numerical investigation of the effect of ALUMINUM porous fins on heat transfer, Heat Transf. Res. 56 (9) (2025) 75–92, https://doi.org/10.1615/HEATTRANSRES.2025057266.
[33] J. Guo, G. Huang, C.N. Markides, Solar harvesting maximisation in spectral-splitting photovoltaic-thermal systems via integrated radiative cooling, Energy 328 (2025), https://doi.org/10.1016/j.energy.2025.136455.
[34] Y. Hu, T. Guo, X. Qu, T. Xu, B.T. Huang, T. Xie, Enhanced passive radiative cooling of cementitious composites for efficient photovoltaic thermal management, Energy Convers. Manag. 344 (2025), https://doi.org/10.1016/j.enconman.2025.120268.
[35] H. Raad, S. Ali, J. Faraj, C. Castelain, K. Chahine, M. Khaled, Enhancing photovoltaic panel efficiency through water-cooling: a parametric comparative evaluation of energetic, economic, and environmental benefits, Unconventional Resources 7 (2025), https://doi.org/10.1016/j.uncres.2025.100208.
[36] B. Jiang, et al., Experimental and numerical investigation of photovoltaic modules with rear spray cooling system, Appl. Therm. Eng. 284 (2026), https://doi.org/10.1016/j.applthermaleng.2025.129035.
[37] S.M. Hussein, M. Jafari, M.T. Shervani-Tabar, S.E. Razavi, Numerical study of the effect of a new cooling method using twisted tape with water on the performance of photovoltaic, Ain Shams Eng. J. 17 (1) (2026), https://doi.org/10.1016/j.asej.2025.103852.
[38] D.H. El-Nagar, M. Emam, A.A. El-Betar, S.A. Nada, Combining active and passive cooling techniques for maximizing power generation and thermal utilization of photovoltaic panels, Renew. Energy 250 (2025), https://doi.org/10.1016/j.renene.2025.123310.
[39] L. Zhang, et al., Experimental study of dual-source heat pump coupled with water-cooled photovoltaic/thermal system, Appl. Therm. Eng. 258 (2025), https://doi.org/10.1016/j.applthermaleng.2024.124758.
[40] M.I.A. Zaini, et al., Innovative water-cooling system for enhanced energy efficiency in photovoltaic-thermal system, Case Stud. Therm. Eng. 70 (2025), https://doi.org/10.1016/j.csite.2025.106137.
[41] M.B. Ben Hamida, A. Khelifa, M. El Hadi Attia, M.M. Abdel-Aziz, Maximizing energy output in PVT systems with triangular-finned solar air collectors, Appl. Therm. Eng. 278 (2025), https://doi.org/10.1016/j.applthermaleng.2025.127247.
[42] M.E.H. Attia, A.E. Kabeel, A. Khelifa, M.M. Abdel-Aziz, R. Sathyamurthy, W.M. El-Maghlany, Optimal air cooling gap width at the top of the PVT: thermal and electrical efficiency analysis, J. Therm. Anal. Calorim. 150 (3) (2025) 2047–2058, https://doi.org/10.1007/s10973-024-13882-2.
[43] M.B. Ben Hamida, A. Khelifa, M.E.H. Attia, M.M. Abdel-Aziz, Optimizing baffle plate inclination in PV cooling systems for maximum heat dissipation, Appl. Therm. Eng. 278 (2025), https://doi.org/10.1016/j.applthermaleng.2025.127387.
[44] M.R. Sharaby, S.W. Sharshir, A.A. ElBahloul, A.W. Kandeal, M. Rashad, Performance evaluation of fixed and sun-tracking photovoltaic systems integrated with spray cooling, Sol. Energy 288 (2025), https://doi.org/10.1016/j.solener.2025.113310.
[45] H. Chen, Y. Liu, Y. Cui, Y. Zhao, T. Hong, J. Ma, Enhanced heat transfer performance of concentrated photovoltaic system through liquid spray cooling, Energy 329 (2025), https://doi.org/10.1016/j.energy.2025.136790.
[46] Y. Zhao, et al., Experimental and numerical investigation of spray cooling based photovoltaic/thermal system: achieving high performance, low cost, and lightweight design, Energy 323 (2025), https://doi.org/10.1016/j.energy.2025.135671.
[47] A. Kabakuş, F. Sönmez, A. Öztürk, Thermal analysis of photovoltaic panel cooled by electrospray using different fluids, Duzce University J. Sci. Technol. 12 (4) (2024) 2271–2282, https://doi.org/10.29130/DUBITED.1492678.
[48] A. Kilic, K. Yakut, M.H. Osta, A.N. Özakin, H. Ghaly, A novel air-assisted spray cooling approach using water–acetone mixtures for enhanced photovoltaic thermal management, Int. J. Heat Fluid Flow 119 (2026), https://doi.org/10.1016/j.ijheatfluidflow.2026.110286.
[49] M.B. Ben Hamida, A. Khelifa, M. El Hadi Attia, M.M. Abdel-Aziz, Parallel flow cooling in hybrid PV/T systems: a computational investigation of air and water integration, Therm. Sci. Eng. Prog. 65 (2025), https://doi.org/10.1016/j.tsep.2025.103973.
[50] Z. Noui, et al., Mini-channel heat sink design for solar photovoltaic cooling: experimental evaluation and performance metrics, Alex. Eng. J. 130 (2025) 85–100, https://doi.org/10.1016/j.aej.2025.09.010.
[51] C. Wang, G. Zhang, Y. Wang, L. Song, B. Liu, Comprehensive optimization of photovoltaic-thermoelectric hybrid systems: experimental analysis of cooling methods, photovoltaic strategies, and their synergistic effects, Sol. Energy 298 (2025), https://doi.org/10.1016/j.solener.2025.113667.
[52] X. Xie, et al., Photovoltaic efficiency enhancement via responsive forced convection driven by radiative-cooling-intensified thermoelectricity, Chem. Eng. J. 521 (2025), https://doi.org/10.1016/j.cej.2025.166606.
[53] T. Ibrahim, et al., 3E parametric analysis of hybrid phase change material cooling techniques for enhanced photovoltaic panel efficiency – a power plant case study, Appl. Therm. Eng. 284 (2026) 129133, https://doi.org/10.1016/j.applthermaleng.2025.129133.
[54] H.L. Azeez, A. Ibrahim, J. Kasprzak, B.O. Ahmed, A.H.A. Al-Waeli, M. Jaber, Economic and environmental feasibility analysis of a photovoltaic thermal system with passive cooling techniques, nanofluid, and phase changing materials, Appl. Therm. Eng. 274 (2025), https://doi.org/10.1016/j.applthermaleng.2025.126782.
[55] J. Li, Z. Hu, Y. Xue, J. Li, H. Zhang, Actively cooling photovoltaic modules and simultaneously yielding freshwater via humidifier-dehumidifiers: performance predictions and regulatory strategies, Appl. Therm. Eng. 268 (2025), https://doi.org/10.1016/j.applthermaleng.2025.125909.
[56] E.C. Rolin, G.R. Thomaz, L.M. Moura, Energy efficiency of thermoelectric photovoltaic systems with thermoelectric cooling: a study based on numerical modeling and parameter identification, Case Stud. Therm. Eng. 76 (2025) 107410, https://doi.org/10.1016/j.csite.2025.107410.
[57] Y. El Alami, et al., Experimental-numerical comparative study of performance and cost-effectiveness of partially- and fully-cooled photovoltaic thermal systems, Case Stud. Therm. Eng. 73 (2025) 106660, https://doi.org/10.1016/j.csite.2025.106660.
[58] M.H. Abdel-Aziz, et al., Optimizing concentrated photovoltaic module efficiency using nanofluid-based cooling, Energy Convers. Manage.: X 26 (2025), https://doi.org/10.1016/j.ecmx.2025.100928.
[59] M.A. Baghal Khosh Naghsh, M.H. Shahverdian, H. Sayyaadi, R. Javadijam, CFD-driven regression models for enhancing photovoltaic performance with nano fluid cooling: economic and environmental analysis, Case Stud. Therm. Eng. 72 (2025), https://doi.org/10.1016/j.csite.2025.106352.
[60] M.E.H. Attia, A.E. Kabeel, A. Khelifa, M. Abdelgaied, M. Arıcı, M.M. Abdel-Aziz, Performance enhancement of PVT modules using bi-fluid (air/CuO–water-based

nanofluid) and fins: energy and exergy analysis, Clean Techn. Environ. Policy 27 (9) (2025) 4541–4559, https://doi.org/10.1007/s10098-025-03149-1.
[61] H. Togun, et al., Advancing photovoltaic thermal (PV/T) systems: innovative cooling technique, thermal management, and future prospects, Elsevier Ltd, 2025, https://doi.org/10.1016/j.solener.2025.113402.
[62] H. Ghaly, K. Yakut, M.H. Osta, Experimental investigation of cube cavity synthetic jet performance on heat sinks, Heat Transf. Res. 56 (18) (2025), https://doi.org/10.1615/HEATTRANSRES.2025058034.
[63] P. Gil, Flow and heat transfer characteristics of single and multiple synthetic jets impingement cooling, Int. J. Heat Mass Transf. 201 (2023), https://doi.org/10.1016/j.ijheatmasstransfer.2022.123590.
[64] V. Arumuru, K. Rajput, R. Nandan, P. Rath, M. Das, A novel synthetic jet based heat sink with PCM filled cylindrical fins for efficient electronic cooling, J. Energy Storage 58 (2023), https://doi.org/10.1016/j.est.2022.106376.
[65] P. Gil, J. Wilk, M. Korzeniowski, Helmholtz resonance frequency of the synthetic jet actuator, Appl. Sci. 11 (12) (2021) 5666, https://doi.org/10.3390/app11125666.
[66] Z. Liu, Q. Yu, Z. Mei, D. Xie, Impingement cooling performance analysis and improvement of synthetic jets for electronic devices 33, 2019, pp. 856–864, https://doi.org/10.2514/1.T5723, 3.
[67] K. Bakirci, Estimation of solar radiation by using ASHRAE clear-sky model in Erzurum, Turkey, Energy Sources, Part A 31 (3) (2009) 208–216, https://doi.org/10.1080/15567030701522534.
[68] J. Siecker, K. Kusakana, B.P. Numbi, A review of solar photovoltaic systems cooling technologies, Renew. Sust. Energ. Rev. 79 (2016) 192–203, 2017, https://doi.org/10.1016/j.rser.2017.05.053.